MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering

Subject: Object-Oriented Programming Using C++ (Paper Code: CIC-211)

PROJECT TECHNICAL DESIGN & IMPLEMENTATION REPORT

Autonomous Landslide & Flash-Flood Early Warning Grid (ALF-EWG)

A High-Performance C++ Object-Oriented IoT Telemetry & Disaster Management System

Project Option: Option 1 (ALF-EWG)	**Standard:** Clean ISO C++14/17 Standard
Course Code: CIC-211 (OOP Using C++)	**Target Domain:** Himalayan Disaster Risk Management
Source File: ALF-EWG.cpp	**Submission Date:** November 2026
Architecture: 4-Unit Syllabus Compliant	**Execution Status:** Verified (Exit Code 0)

1. Domain & System Context

In vulnerable mountainous regions such as the Himalayan belt, extreme precipitation and seismic instability trigger sudden catastrophic landslides and flash floods, crippling transportation corridors, electric grids, and residential settlements. The Autonomous Landslide & Flash-Flood Early Warning Grid (ALF-EWG) is designed as a mission-critical edge-and-backend telemetry ingestion system deployed across remote solar-powered IoT monitoring stations.

ALF-EWG autonomously collects multi-modal sensor streams (pore pressure, soil moisture, geophone vibrations, tiltmeters, and river-level sonar gauges), continuously calculates hazard indexes in real time, issues multi-tier regional siren alerts, executes priority-based emergency evacuation routing, and atomically logs all incidents to persistent binary disk storage with transactional rollback support.

Key Architectural Principles:
- ***Strict Encapsulation:*** *Separation of raw sensor drivers from processing alerts via dedicated namespaces.*
- ***Zero Memory Leaks:*** *RAII resource management across dynamically allocated sensor circular buffers.*
- ***Ambiguity-Free Multiple Inheritance:*** *Virtual base classes resolving the classical C++ Diamond Problem.*
- ***Exception-Safe Persistence:*** *Atomic file logging supporting commit/rollback points via stream position manipulation.*

2. System Architecture & Object-Oriented Design

The software architecture enforces clean separation between classification hierarchies and component containment, directly addressing the core competencies of the CIC-211 curriculum.

2.1 Virtual Base Classes & Diamond Problem Resolution

When an autonomous IoT probe inherits both solar harvesting capabilities (SolarPoweredNode) and wireless LoRa communication telemetry (WirelessLoRaTransceiver), both branches derive from the common root hardware abstraction DisasterTelemetryDevice. Without virtual inheritance, the leaf probe AutonomousHybridProbe would inherit duplicate copies of the root device state (serial number, calibration status, base geocoordinates), resulting in compile-time ambiguity.

```
+----------------------------+ | DisasterTelemetryDevice | <-- Root (Virtual Base Class) | (serial, lat,
lon, calibrate)| +----------------------------+ / \ / \ (virtual inheritance) / \
+------------------------+ +--------------------------+ | SolarPoweredNode | | WirelessLoRaTransceiver | |
(harvestSolar, energy) | | (freq_MHz, transmitPacket)| +------------------------+
+--------------------------+ \ / \ / (multiple inheritance) \ / +----------------------------------+ |
AutonomousHybridProbe | <-- Leaf Probe (Unified Subobject) | (readTelemetry, evaluateThreshold) |
+----------------------------------+
```

Virtual Base Resolution Mechanism: By declaring class SolarPoweredNode : public virtual DisasterTelemetryDevice and class WirelessLoRaTransceiver : public virtual DisasterTelemetryDevice, the C++ runtime delegates construction of the shared subobject directly to the most-derived leaf constructor (AutonomousHybridProbe). This guarantees exactly one copy of the root state in memory.

2.2 Containership (Aggregation) vs. Inheritance

To satisfy the design principle of 'favoring composition over inheritance', PowerSubsystem is modeled as an independent, encapsulated class and aggregated directly inside BaseSensorNode (containership). A sensor node *has a* power unit rather than *is a* power unit. This allows battery discharge cycles, solar recharging, and low-power threshold evaluations to be decoupled from sensor telemetry logic.

3. Technical Requirements Mapping Across Syllabus

Every technical specification detailed in the CIC-211 Project Assignment has been rigorously fulfilled in the codebase. The following sections describe the mechanism, theoretical significance, and concrete implementation.

3.1 Unit I: Core Foundations & Call Semantics

Requirement	Implementation & Technical Role
Distinct Namespaces	DisasterCore::Sensors and DisasterCore::Alerts partition mathematical processing from notification state dispatch, preventing identifier collisions.
Inline Mathematical Risk Functions	computeHazardIndex() algorithms are declared inline to eliminate stack-frame call overhead during microsecond-level telemetry processing loops.
Function Overloading	computeHazardIndex() is overloaded with 3 specialized signatures: (1) rainfall rate + slope angle; (2) soil saturation + vibration + river sonar; (3) all 4 parameters combined.
Call by Reference	transformGeoCoordinates(double& lat, double& lon, double dLat, double dLon) accepts references to avoid copying coordinate structs during high-frequency GPS transforms.
Return by Reference	accumulateTelemetry(double& accumulator, double reading) returns a lvalue double& to allow chained state mutations and immediate cumulative state tracking without temporary copies.
Friend Class Audit	TelemetryAuditor is declared as a friend class of SensorStation, granting access to private telemetry buffers and calibration offsets without violating encapsulation via public getters.

3.2 Unit II: Encapsulation, Dynamic Memory & Operator Overloading

Requirement	Implementation & Technical Role
Entities & Const Correctness	SensorStation, EarlyWarningSiren, and TelemetryBuffer maintain strict information hiding. Member accessors (average(), getId()) are declared const.
Static Data & Member Functions	SensorStation::activeCount_ and SensorStation::globalEmergencyLevel_ along with EarlyWarningSiren::totalSirens_ track distributed network state across all instances.
Dynamic Constructors & Deep Copy	TelemetryBuffer dynamically allocates double* readings_ in its parameterized constructor. Explicit copy constructor and copy assignment operator perform deep copies, preventing double-free corruption.
Operator Overloading (+)	operator+(SensorStation& rhs) aggregates two or more sensor stations into a regional zone vector (std::vector).
Operator Overloading (>)	operator>(const SensorStation& rhs) provides natural relational comparison, prioritizing stations with higher hazard index values for triage ordering.
Operator Overloading (<<, >>)	operator<< outputs tabular row formatting for console and file streams; operator>> extracts telemetry streams into stations and buffers.

3.3 Unit III: Inheritance, Containership & Polymorphism

Requirement	Implementation & Technical Role
Abstract Base Class	BaseSensorNode declares pure virtual functions readTelemetry() = 0 and evaluateThreshold() = 0, enforcing a strict polymorphic interface.
Virtual Base Classes	DisasterTelemetryDevice serves as virtual base class for SolarPoweredNode and WirelessLoRaTransceiver, resolving the diamond inheritance in AutonomousHybridProbe.
Containership (Aggregation)	BaseSensorNode aggregates an independent PowerSubsystem instance (battery level, solar charging) rather than using deep inheritance.
Runtime Dynamic Dispatch	RegionalSweepEngine manages std::vector base pointers. Iterative sweep cycles execute dynamic method dispatch via vtable/_vptr bindings at runtime.

3.4 Unit IV: Streams, Persistence, Exception Safety & STL

Requirement	Implementation & Technical Role
Custom Exception Hierarchy	SensorFailureException, TelemetryChecksumMismatch, and ThresholdOverflowException inherit from std::exception and provide diagnostic what() overrides.
Persistent Binary Logging	IncidentLogger serializes POD IncidentRecord structs using binary std::fstream (std::ios::binary | std::ios::in | std::ios::out).
Transactional Rollbacks	beginTransaction() records commit offsets. If corrupt records occur, rollback() reposition file pointers using seekp() and seekg() to mark invalid entries atomically.
STL Containers & Algorithms	std::vector and std::map store nodes and hazard registries. std::sort and std::for_each perform ranking and pipeline dispatching.
Custom Functors	HazardPriorityComparator (descending hazard sort) and AlertDispatcher (conditional route filtering) provide functional dispatch objects.
Custom Stream Manipulators	alertHeader, alertFooter, and parameterized ColFmt encapsulate tabular column formatting for clean console reporting.

4. Verification & Live Execution Output

The compiled binary (alf_ewg.exe) was executed on the target environment. The entire lifecycle—from station booting to exception handling, polymorphic sweep, binary persistence, and evacuation dispatch—executed synchronously with exit code 0.

```
========================================================================== ALF-EWG: Autonomous Landslide &
Flash-Flood Early Warning Grid ============================================================================== ---
Booting Sensor Stations --- [Alert] Ingestion warning: ThresholdOverflowException: value=92.000000 exceeded
threshold=90.000000 --- Activating Early Warning Sirens --- [SIREN #1 @ Kedarnath-Village] ACTIVATED -> RED
(Danger) [SIREN #2 @ Badrinath-Gate] ACTIVATED -> YELLOW (Watch) [SIREN #3 @ Chamoli-Township] ACTIVATED ->
CRITICAL (Evacuate) Network: 3 sirens total, 3 active | Stations online: 3 --- Regional Station Status ---
========================================================================== ID Station HazardIndex AlertLevel
Active ---------------------------------------------------------------------- 1 Kedarnath-North 67.00 RED
(Danger) YES 2 Badrinath-East 31.85 YELLOW (Watch) YES 3 Chamoli-Valley 92.00 CRITICAL (Evacuate) YES
========================================================================== --- Hazard Index Calculations ---
computeHazardIndex(rainfall=85, slope=42) = 67.800 computeHazardIndex(soil=78, vibr=5.5, river=3.2) =
33.925 computeHazardIndex(rain=70, slope=35, sat=65, v=4) = 45.300 Geocoord transform -> (30.318, 78.034)
Telemetry accumulator -> 117.400 --- Calibrating Autonomous Hybrid Probes (Diamond Hierarchy) --- [Device
AHP-KDN-01] Calibrated at (30.7340, 79.0660) [Device AHP-BDR-02] Calibrated at (30.7440, 79.4930) [Device
AHP-CHM-03] Calibrated at (30.4000, 79.3500) [Solar AHP-KDN-01] Harvested 10.80 Wh today --- Cyclical Sensor
Sweep (Polymorphic vtable Dispatch) --- Node[101] HazardIdx=51.90 -> ORANGE (Warning) Node[102]
HazardIdx=37.10 -> YELLOW (Watch) [LoRa @867.00MHz SF=10.00] Pkt#1: ALERT! HazardIdx=61.440000 Node[103]
HazardIdx=61.44 -> RED (Danger) * THRESHOLD TRIGGERED --- Incident Logging (Binary fstream) --- [Logger]
Written: Station#1 HazardIdx=67.0 Alert=RED (Danger) [Logger] Written: Station#2 HazardIdx=31.9 Alert=YELLOW
(Watch) [Logger] COMMIT at position 208 [Logger] Written: Station#99 HazardIdx=999.9 Alert=CORRUPT Corrupt
write detected -> rolling back... [Logger] ROLLBACK: marked 1 record(s) as rolled-back (seekp/seekg)
Verified records on disk: 2 Station#1 HazardIdx=67.0 Alert=RED (Danger) Time=2026-11-11T15:00:00 Station#2
HazardIdx=31.9 Alert=YELLOW (Watch) Time=2026-11-11T15:05:00 --- Evacuation Route Dispatch --- [DISPATCH]
Chamoli-Zone-C Hazard=91.0 Pop=2500 Via: SH-13 via Gopeshwar [DISPATCH] Kedarnath-Zone-A Hazard=85.0
Pop=1200 Via: NH-58 via Gaurikund [DISPATCH] Badrinath-Zone-B Hazard=62.0 Pop=800 Via: NH-7 via Joshimath
[DISPATCH] Rudraprayag-Zone-D Hazard=45.0 Pop=600 Via: NH-58 via Tilwara Zones dispatched: 4 / 5
========================================================================== ALF-EWG System shutdown complete.
All heap memory released. ==============================================================================
```

5. Memory Management & Exception Safety Analysis

RAII & Heap Integrity: All dynamically allocated resources in TelemetryBuffer (raw double* buffer arrays) and SensorStation are managed via strict RAII protocols. Parameterized dynamic constructors allocate buffers, deep copy constructors and copy assignment operators duplicate buffer contents to isolate object lifetimes, and explicit destructors call delete[], preventing dangling pointers and memory leaks upon scope termination.

Exception Guarantees: Custom exceptions derived from std::exception (such as ThresholdOverflowException) provide the strong exception safety guarantee. When emergency limits are breached, station states are updated safely before control transfers to the caller's catch block, allowing continuous operational monitoring.

6. Project Submission Summary

Project Option:	Option 1: Autonomous Landslide & Flash-Flood Early Warning Grid
Source File:	ALF-EWG.cpp (Tested & compiled with MinGW g++)
Executable Binary:	ALF-EWG.exe (Exit code: 0, clean termination)
Persistence Artifact:	incidents.bin (Validated binary stream serialization)
Academic Subject:	Object-Oriented Programming Using C++ (CIC-211)
Institution:	Maharaja Agrasen Institute of Technology (MAIT)